DISSERTATION INTRODUCTION

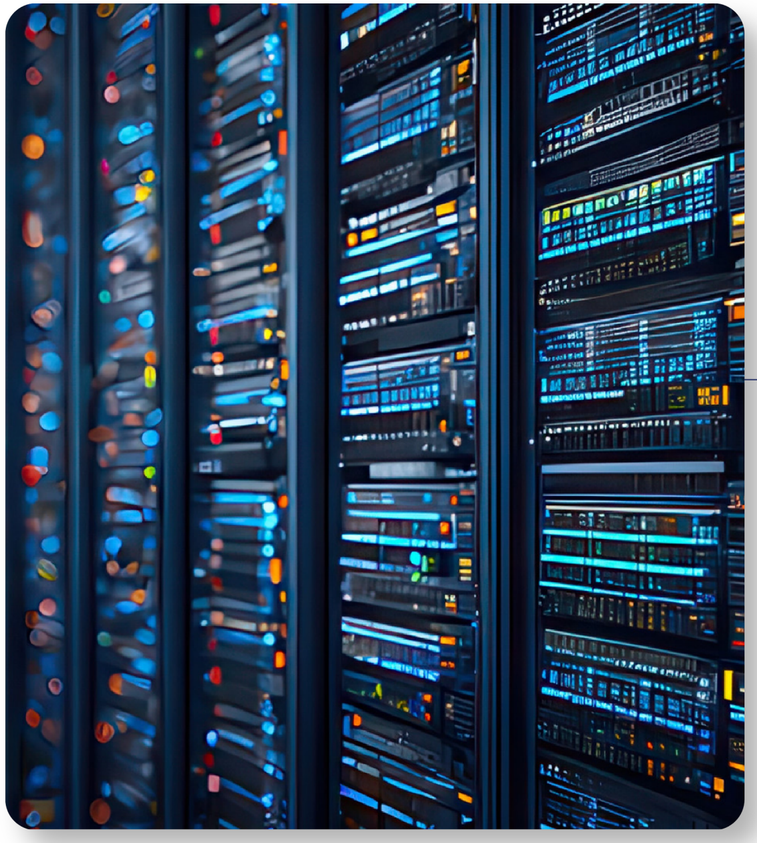
Hallucination Across various Retrieval Augmented Generation (RAG) Strategies

A comparative study on how different RAG architecture influence LLM hallucinations.

ANKIT MAHTO - 13454

INTRODUCTION

- In modern times, advancement in Artificial intelligence has given birth to large language models have been revolutionary across all aspects of human life.
- Large language models are being adopted in various fields such as medical, banking, retail, corporate, academia...
- However, these LLMs have some underlying issues, most notable being LLM hallucinations.
- This is where RAG comes in, in recent years, RAG has proven to be a great method for mitigating LLM hallucinations along with other problems.
- This research paper will compare and contrast different architectures of Retrieval-Augmentation Generation to find correseponding hallucination rates in large language models (LLMs).
- This research paper will be descriptive/exploratory in nature.



BACKGROUND

1

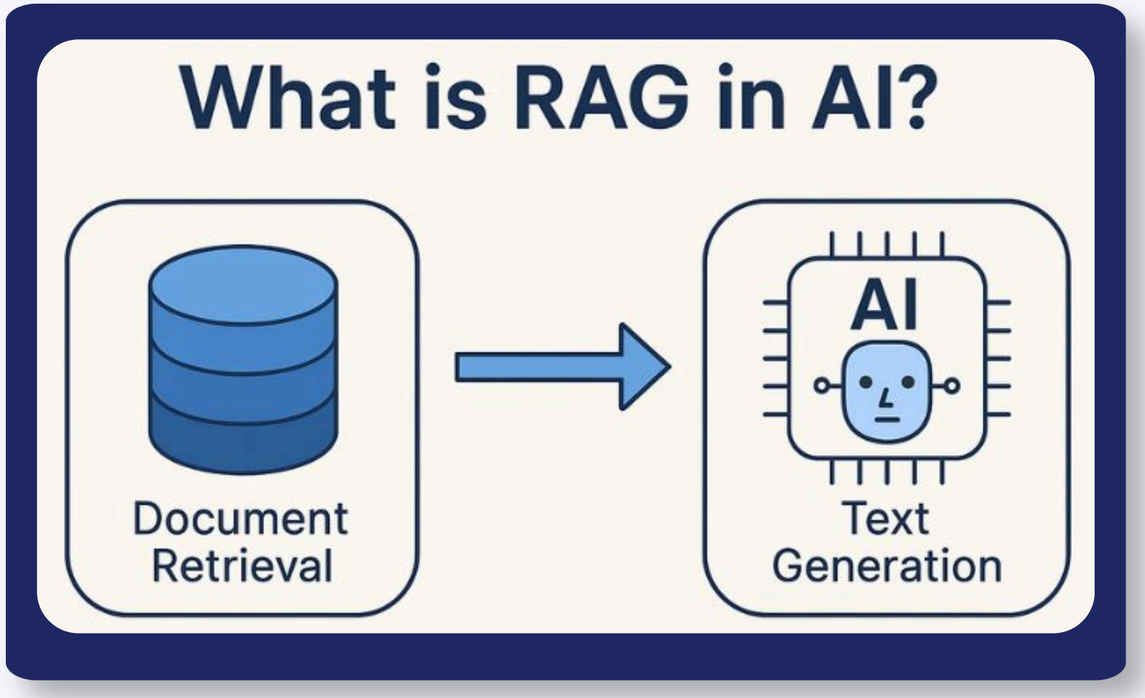
What is retrieval-augmentation generation (RAG)

RAG is a AI architecture by which a LLM model can access private data.

2

Why do we need RAG?

- A LLM is pretrained models that predicts the ouput based on the parametric knowledge (pretrained knowledge).
- A LLM by default doesn’t have access to outside knowledge
- This leads to several issues, which we will see further



Problems RAG solves with plain LLM

Common issues encoutered when dealing with vanilla LLM

1

Knowledge cut-off

LLMs don’t have the knowledge of newere data which they werent trained on. so they can’t answer questions related to recent events.

2

Lacking private data

LLMs dont have access of the personal data. Therefore, they can’t answer questions related to personal data

3

Expensive training

Training LLMs with newer data takes time and cost. RAG solves this by feeding newer knowledge to LLMs directly

4

Hallucinations

When LLMs don’t have the required data, they hallucinate and make up rnadom information and reply confidently. This is called hallucination effect

These are some of the core problems that RAG solves effectively. But for the sake of this study we will focus on hallucination part.

Basic architecture of RAG

RAG Components

Indexing

Retrieval

NOTE: for the sake of this study, we will only only consider two major components of RAG as these are where most changes and comparisons would be made. Other parts such as Augmentation and generation will mostly remain constant

RAG Taxonomy

Retrieval decides what the model even gets to see

1

Query

A question comes in, needing information beyond the model's parameters.

2

Retrieve

An index (vectors, keywords, or a graph) pulls candidate passages from a corpus.

3

Augment

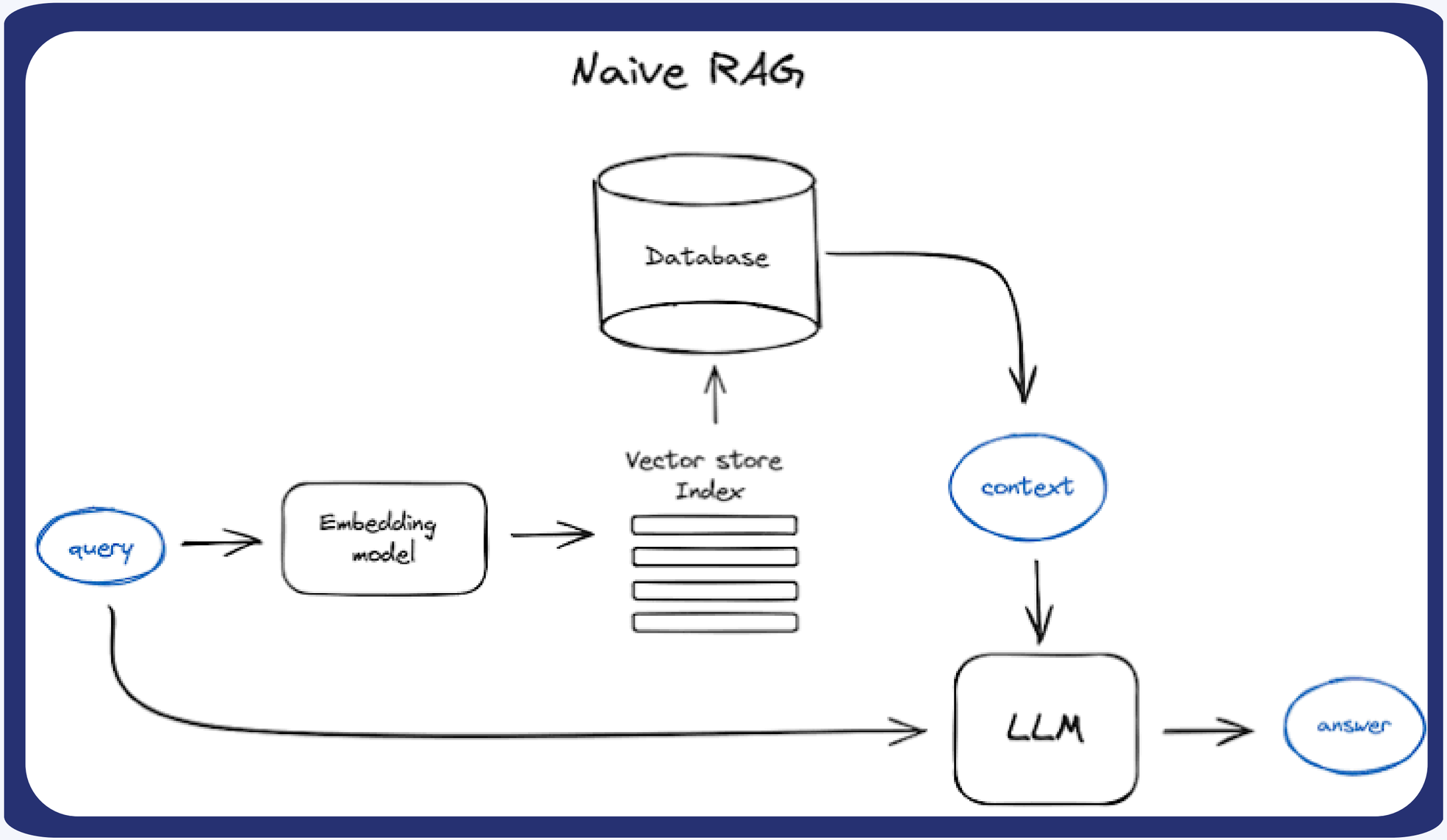
Retrieved passages are inserted into the model's context window.

4

Generate

The LLM writes an answer — ideally grounded in what it was handed.

Naive RAG Architecture



Other RAG Architectures

Naive RAG

Advanced RAG

Graph RAG

Modular RAG

- Self-RAG
- Corrective RAG
- Agentic RAG
- Adaptive RAG

Literature review

S.no.	Paper	Publisher	Subject	Outcome
1	Maximizing RAG efficiency: A comparative analysis of RAG methods	Cambridge University Press: 30 October 2024		
2	EvaRAG: Evaluating Advanced RAG Techniques With Indexing and Distance Metrics	IEEE		
3	TYPED-RAG: Type-Aware Decomposition of Non-Factoid Questions for Retrieval-Augmented Generation	Association for Computational Linguistics		
4	Re2G: Retrieve, Rerank, Generate	Association for Computational Linguistics		
5	A Survey on RAG with LLMs	Science Direct		
6	A survey on retrieval-augmentation generation (RAG) models for healthcare applications	Springer	Compares types of rag in Medical Field	
7	A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications	IEEE	Compares different retreival types	
8	Optimising clinical information extraction: a comparative study of retrieval-augmented generation techniques in clinical notes	Science Direct	Compares different retreival types in medical field	
9	Evaluating Sparse and Dense Retrieval in Retrieval-Augmented Generation Systems: A Study	Association for computing machinery (ACM)	Compares different retreival types	
10				

Methodology

Proposed mehtodology for this study



Datasets

Datasets that can be used for retrieval

Medical

- MedRAG
- Mirage
- pubMedQA

Law

- Legalbench-RAG
- SC Judgements
- Legal RAG bench

Corporate

- EnterpriceRAG bench
- RAG multi corpus
- Hugging face datasets

Science

- SciQ
- RAGCare-QA
- Open RAG bench

LLM Models

Top Open source models

KIMI K3

Parameters: 2.8 Trillion
Context length: 1 million token
Disk size: 1.56 TB

GLM 5.2

Parameters: 753 Billion
Context length: 1 million token
Disk size: 1.51 TB

Gemma 4 26B A4B

Parameters: 26 Billion
Context length: 256K token
Disk size: 48 GB

Deepseek V4 Pro

Parameters: 1.6 Trillion
Context length: 1 million token
Disk size: 893 GB

Qwen 3.8-Max

Parameters: 2.4 Trillion
Context length: 1 million token
Disk size: 1.2 TB

Llama 4 Maverick

Parameters: 400 Billion
Context length: 1 million token
Disk size: 800 GB

Min-Max M3

Parameters: 428 Billion
Context length: 1 million token
Disk size: 128 GB

Things to consider

Changes can be applied to different components

Core Architecture

- Naive RAG
- Advanced RAG
- Modular RAG
- Agentic RAG

Retrieval mechanism

- Sparse (BM25)
- Dense (embeddings)
- Hybrid (fused)

Knowledge structure

- Flat / chunked
- Hierarchical (RAPTOR)
- Graph RAG

Control flow/behavior

- Self-RAG
- Corrective RAG
- FLARE
- Adaptive RAG

Things to consider

Changes can be applied to different components

RAG EVALS

Score the whole pipeline

- Context precision — are retrieved chunks relevant?
- Context recall — was everything needed retrieved?
- Faithfulness — does the answer match retrieved context?
- Answer relevance — does it address the query?

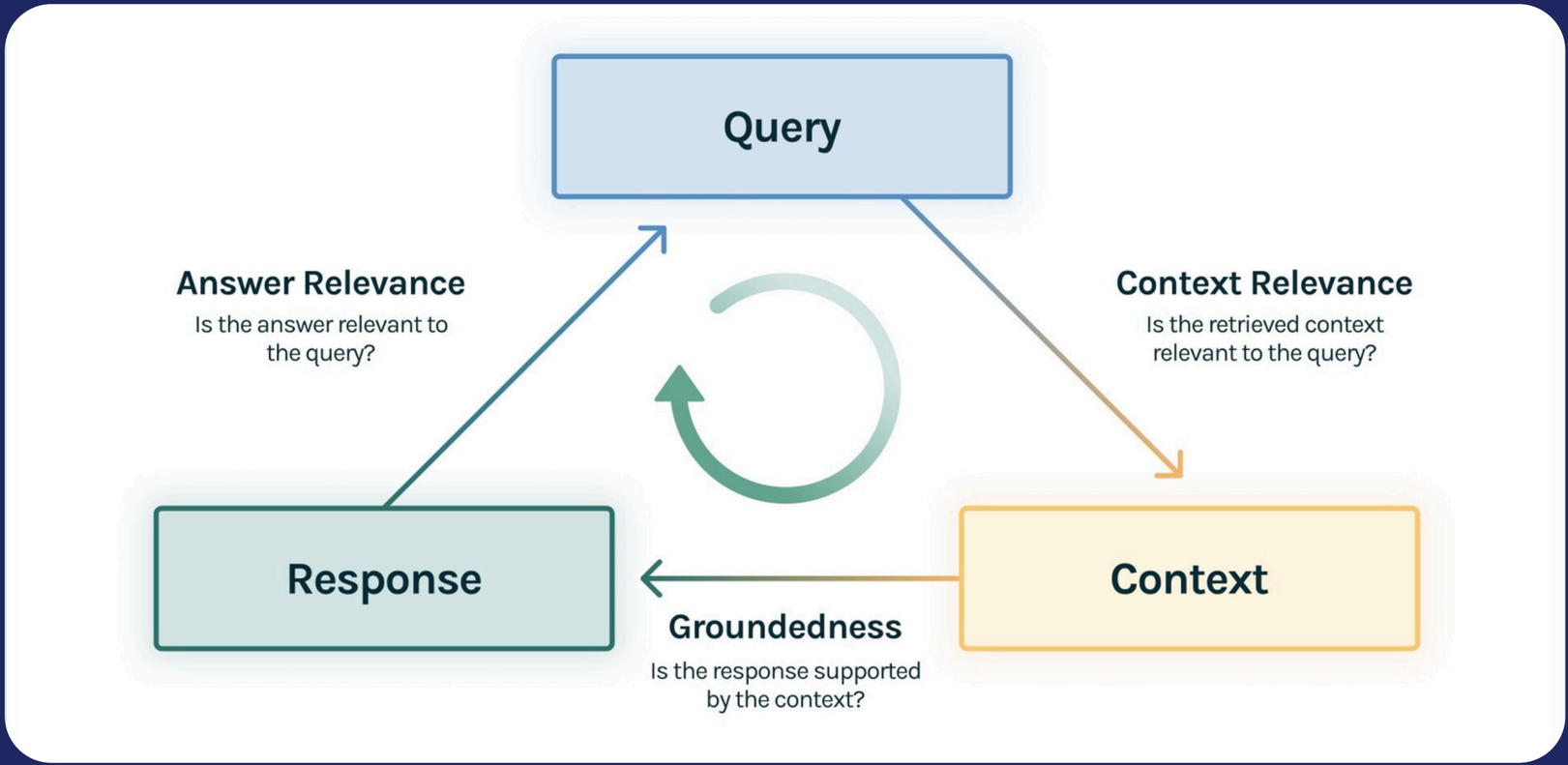
HALLUCINATION EVALS

Score the output on its own terms

- TruthfulQA — common-misconception factuality
- HaluEval — paired correct / hallucinated samples
- SelfCheckGPT — consistency across samples
- RAGTruth — word-level hallucination spans in RAG

Evals

Evaluation Metric: RAG Triad



NOTE: Hallucination on in itself cannot be measured by just single metric, thats why we will be looking at multiple factors to judge the outcome. The best way to judge this is by using RAG traid. It measures the out put based on 3 metrics: Context Relevance, Answer Relevance and groundedness. It does 2 test before output generation and 2 after the generation.

Evals

Potential Eval frameworks

RAGAS

DeepEval

LangSmith

Thank you